Market-Implied Treasury Supply News

High-frequency Treasury announcements, local projections, and robustness checks

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What This Project Shows

- Builds a source-audited Treasury event database from QRA policy statements and coupon-auction announcements.
- Converts exact-minute Treasury release windows into a monthly market-implied supply-news proxy.
- Estimates dynamic yield-curve responses with lag-augmented local projections.
- Separates strong reduced-form evidence from weaker expected-size LP-IV interpretation.
- Uses placebo windows, matched non-events, narrow windows, Databento futures, and mechanism states to discipline the claim.

One-Slide Answer

- A one-unit expanding-standardized Treasury supply-news proxy raises same-month yield changes by about 3.2–4.1 bps.
- Later monthly yield-change coefficients turn negative for some maturities; this is not by itself a cumulative level-reversal claim.
- Calendar matched non-events remain large: matched absolute reactions are 83.2 percent of true release reactions.
- Volatility-regime matching is implemented as a coverage audit, not headline evidence, because current pseudo-event windows lack usable intrawindow returns.
- LP-IV expected-size evidence is weak and sign-conflicted, so the headline remains reduced form.

Research Design

Object

$$z_t^{LEV} = \frac{Z_t^{LEV} - \hat{\mu}_{t-1}}{\hat{\sigma}_{t-1}}, \qquad Z_t^{LEV} = \sum_{g \in t} R_g^{LEV}.$$

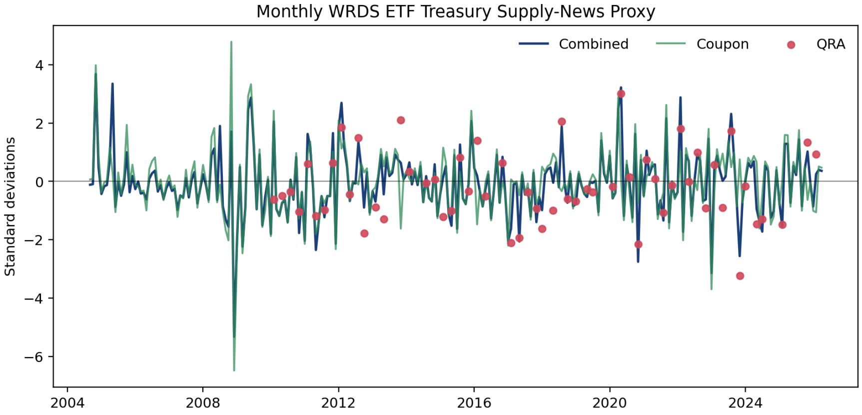
- g indexes exact Treasury information-release timestamps.
- R_g^{LEV} is the high-frequency Treasury ETF level reaction.
- Standardization uses prior monthly observations only.
- The estimand is a response to market-implied supply news, not a dollar issuance multiplier.

Event and Market-Reaction Coverage

Metric	Value
Release groups	615
Groups with level proxy	608
Severity 0 groups	442
Severity 0–1 baseline groups	615
Ticker reactions	3,868
Finite ticker reactions	3,637

Release groups prevent multi-security coupon announcements from duplicating the same ETF return.

Shock Series



Why the Claim Is Conservative

What is supported

- Exact source-audited event timing.
- Treasury ETF market reactions.
- Direct LP responses to the proxy.
- Robustness across windows, lags, controls, and futures checks.

What is not claimed

- A pure structural Treasury supply shock.
- A dollar-per-billion issuance multiplier.
- A clean LP-IV design from expected-size surprises.
- A computed volatility-regime placebo when intrawindow coverage is inadequate.

Matched Non-Event Evidence

Diagnostic	Value	Interpretation
Matched groups	550	Calendar pseudo-events
Ticker-window rows	7,550	Same clock-time design
Matched / true abs. reaction	0.832	Large noise component
True / matched signal-to-noise	1.202	True windows larger, not clean
Signed correlation	-0.026	No same-signed replication

This supports a noisy-proxy interpretation rather than structural-shock language.

Volatility-Regime Coverage Audit

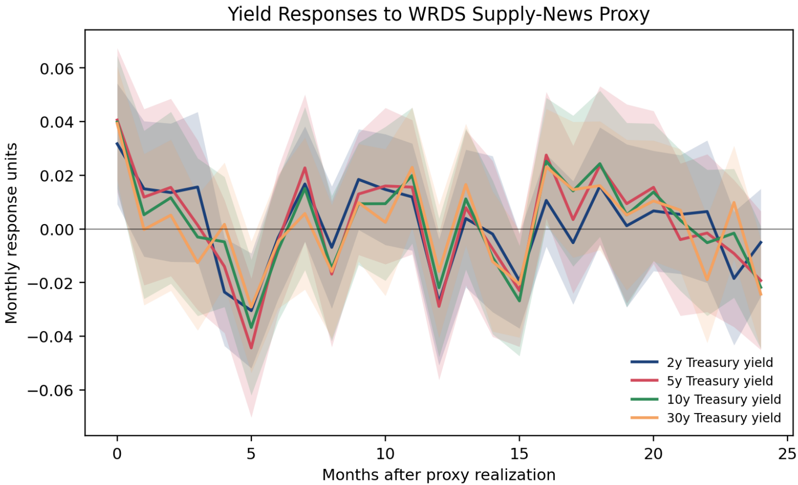
- Intended design matches pseudo-events on weekday, clock time, lagged VIX tercile, and lagged rate-volatility tercile.
- Terciles use expanding-window cutoffs through $t-1$, avoiding full-sample look-ahead.
- Current reusable windows have only one minute per accepted ticker-window.
- A return computed from that panel would be mechanical, so the result is appendix-gated as a coverage audit.

Selected Yield Responses

Outcome	$h = 0$ response	90% CI lower	90% CI upper
2y yield	0.0317	0.0092	0.0542
5y yield	0.0406	0.0137	0.0674
10y yield	0.0398	0.0151	0.0645
30y yield	0.0393	0.0181	0.0605

Entries are percentage-point monthly yield changes; multiply by 100 for basis points.

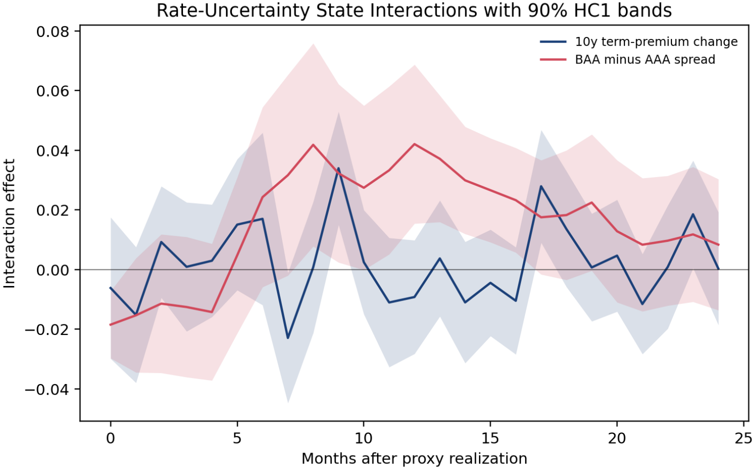
Yield-Curve IRFs



Robustness Map

- Lag-length and factor-control sensitivity checks preserve the main yield-response signs.
- Severity-zero screens test whether minor confounds drive the results.
- Narrow coupon windows reduce contamination risk but can discard delayed price discovery.
- Databento futures provide independent 2011–2026 market-reaction corroboration, not a splice into the WRDS ETF baseline.

Mechanism: Rate Uncertainty



Rate uncertainty is the clearest main-text mechanism state; dealer and positioning screens remain appendix diagnostics.

LP-IV Placement

- Expected-size variables are valuable interpretation layers.
- The combined expected-size first stage is weak and sign-conflicted under the announced-minus-expected convention.
- Anderson–Rubin intervals often hit grid boundaries.
- The paper therefore reports LP-IV readiness as appendix/exploratory evidence after the direct LP results.

Tooling and Reproducibility

- Python pipelines for Treasury event construction, WRDS TAQ reaction extraction, shock validation, LP/LP-IV estimation, mechanism LPs, and paper outputs.
- Data interfaces and audits for WRDS, Databento, LSEG, FRED, Cboe, CFTC, SEC N-PORT, NY Fed/OFR, TRACE/FISD, and Treasury source material.
- Tests verify event coverage, shock construction, validity diagnostics, LP outputs, and appendix gates.
- Public repo keeps code and artifacts, not raw licensed data or credentials.

Takeaway

Core claim

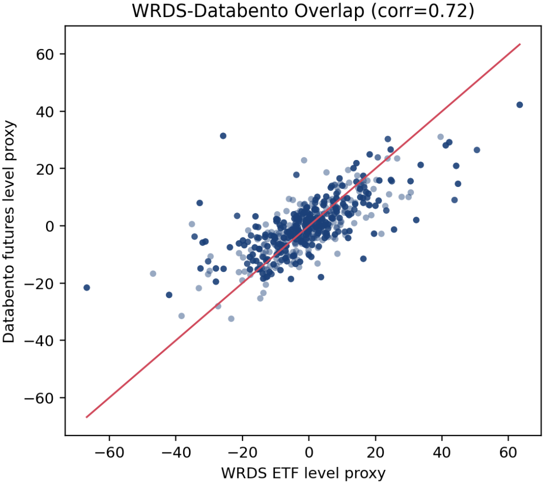
Treasury announcement windows contain market-relevant supply news, and monthly yield changes respond sharply on impact.

- The design is novel because it combines exact source timing, high-frequency Treasury ETF reactions, and conservative validation gates.
- The project is strongest as a disciplined market-implied proxy design.
- The paper avoids overclaiming structural identification when placebo and LP-IV diagnostics do not justify it.

Backup: Narrow Windows

- Baseline windows: QRA $[-10, +30]$, coupon $[-5, +20]$.
- Main narrow check: coupon $[-5, +15]$.
- Stress check: coupon $[-2, +10]$.
- Tighter windows reduce contamination risk but can miss delayed price discovery.

Backup: Databento Futures



Backup: What Would Strengthen the Paper

- A complete pre-event tenor-level auction-size consensus panel.
- Exact CTD mapping for futures yield-equivalent interpretation.
- Event-window Treasury option implied-volatility shocks.
- Exact daily historical ETF holdings and durations.